Lakshya JEE 2.0 (2025)

Physics

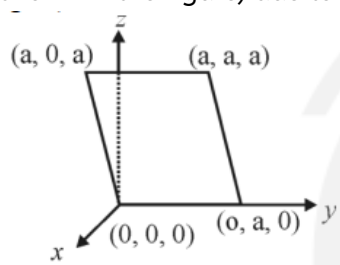
DPP: 8

Electric Charges and Fields

Q1 If the electric field given by $(5\hat{i}+4\hat{j}+8\hat{k})$, the electric flux through a surface of area 20 unit lying in the $Y-Z$ plane will be

(A) 100 unit (B) 80 unit
(C) 180 unit (D) 20 unit

Q2 Consider an electric field $E=E_0\hat{i}$ where E_0 is constant. The flux through the shaded area (as shown in the figure) due to this field is

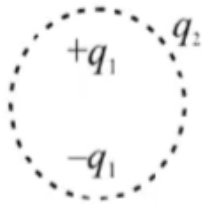


(A) $2E_0a^2$
(B) $\sqrt{2}E_0a^2$
(C) E_0a^2
(D) $\frac{E_0a^2}{\sqrt{2}}$

Q3 The electric flux through a closed surface area S enclosing charge Q is ϕ. If the surface area is doubled, then the flux is

(A) 2ϕ
(B) $\phi/2$
(C) $\phi/4$
(D) ϕ

Q4 Consider the charge configuration and a spherical Gaussian surface as shown in the figure. When calculating the flux of the electric field over the spherical surface, the electric field will be due to:



(A) q_2
(B) Only the positive charges
(C) All the charges
(D) $+q_1$ and $-q_1$

Q5 There exists an electric field of $1\ \mathrm{N/C}$ along y-direction. The flux passing through the square of side $1\ \mathrm{m}$ placed in xy plane inside the electric field is

(A) $1.0\mathrm{Nm^2C^{-1}}$
(B) $10.0\mathrm{Nm^2C^{-1}}$
(C) $2.0\mathrm{Nm^2C^{-1}}$
(D) Zero

Q6 A cylinder of radius R and length L is placed in a uniform electric field E parallel to the cylinder axis. The total flux for the surface of the cylinder is given by

(A) $2\pi R^2E$
(B) $\pi R^2/\mathrm{E}$
(C) $\left(\pi R^2+\pi R^2\right)/\mathrm{E}$
(D) Zero

Q7 An electric charge q is placed at the centre of a cube of side a. The electric flux through one of its faces will be

(A) $\frac{q}{6\varepsilon_0}$
(B) $\frac{q}{\varepsilon_0a^2}$

(C) $\frac{q}{4\pi\varepsilon_0 a^2}$
(D) $\frac{q}{\varepsilon_0}$

Q8 For a given surface the Gauss's law is stated as $\oint E \cdot ds = 0$. From this we can conclude that
(A) E is necessarily zero on the surface
(B) E is perpendicular to the surface at every point
(C) The total flux through the closed surface is zero
(D) The flux is only going out of the surface

Q9 A cube of side l is placed in a uniform field E, where $E = E\hat{i}$. The net electric flux through the cube is
(A) Zero
(B) $l^2 E$
(C) $4l^2 E$
(D) $6l^2 E$

Q10 Eight dipoles of charges of magnitude e are placed inside a cube. The total electric flux coming out of the cube will be
(A) $\frac{8e}{\varepsilon_0}$
(B) $\frac{16e}{\varepsilon_0}$
(C) $\frac{e}{\varepsilon_0}$
(D) Zero

Answer Key

Q1	(A)	Q6	(D)
Q2	(C)	Q7	(A)
Q3	(D)	Q8	(C)
Q4	(C)	Q9	(A)
Q5	(D)	Q10	(D)

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